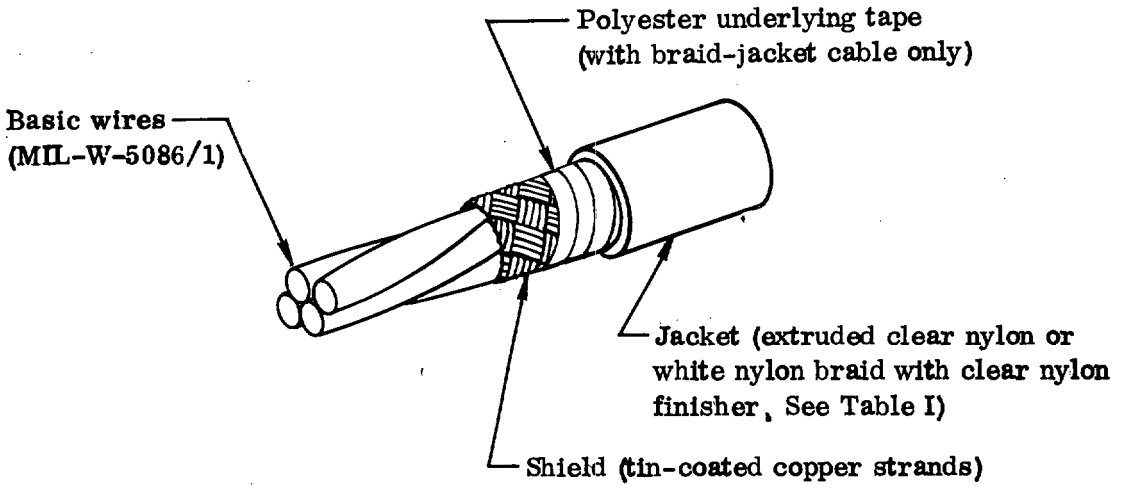
# CABLE, ELECTRIC, AEROSPACE VEHICLE, MIL-W-5086/1 BASIC WIRES, COPPER SHIELD, NYLON JACKET, 600-VOLT, 105° C

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the issues in effect of Specification MIL-C-7078 and Specification Sheet MIL-W-5086/1.




SHIELDED JACKETED CABLE

REQUIREMENTS:
CONSTRUCTION DETAILS: See above Figure and Table I
VOLTAGE RATING: 600 Volts (rms)
TEMPERATURE RATING: 105° C (221° F) max conductor temperature
COLD BEND: Required for extruded-jacket cable only
WET DIELECTRIC TEST AFTER COLD BEND: Not required
THERMAL SHOCK TEST: Required for extruded-jacket cable only
HEAT RESISTANCE: Required for extruded-jacket cable only. Test temperature 130 ±2° C (266 ±3.6° F). Supplementary wet dielectric test not required
JACKET FLAWS (SPARK TEST): 1000 Volts (rms)
DRY DIELECTRIC: 1500 volts (rms)

FSC 6145



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PART NUMBER: Part numbers in this specification sheet are coded as in the following example:

| M7078/3 | - | 22 | - | 1 |
|---|---|---|---|---|
| specification sheet number | | size number of basic wires | | quantity of conductors (basic wire) in cable |

TABLE I

| Cable part no. | Gage of shield strands (AWG) | Thickness of extruded jacket 1/ (in.)(min) | Major diameter of shielded jacketed cable (in.)(max) | Weight of shielded jacketed cable (lb/1000 ft) | |
|---|---|---|---|---|---|
| | | | | (nom.) 2/ | (max) |
| M7078/3-22-1 | 36 | .005 | .112 | 10.2 | 11.4 |
| M7078/3-22-2 | ↓ | .006 | .186 | 18.3 | 20.3 |
| M7078/3-22-3 | ↓ | .006 | .197 | 23.5 | 26.1 |
| M7078/3-22-4 | ↓ | .006 | .216 | 29.2 | 32.5 |
| M7078/3-22-5 | ↓ | .007 | .238 | 35.7 | 39.7 |
| M7078/3-22-6 | ↓ | .007 | .260 | 41.6 | 46.2 |
| M7078/3-22-7 | ↓ | .007 | .260 | 46.0 | 51.1 |
| M7078/3-20-1 | 36 | .005 | .122 | 12.4 | 13.9 |
| M7078/3-20-2 | ↓ | .006 | .206 | 23.1 | 25.4 |
| M7078/3-20-3 | ↓ | .006 | .219 | 30.4 | 33.5 |
| M7078/3-20-4 | ↓ | .007 | .242 | 38.5 | 42.3 |
| M7078/3-20-5 | ↓ | .007 | .265 | 47.0 | 51.7 |
| M7078/3-20-6 | ↓ | Braid | .287 | 55.1 | 59.3 |
| M7078/3-20-7 | ↓ | ↓ | .287 | 61.5 | 66.1 |
| M7078/3-18-1 | 36 | .005 | .132 | 15.9 | 17.1 |
| M7078/3-18-2 | ↓ | .007 | .228 | 29.7 | 31.7 |
| M7078/3-18-3 | ↓ | .007 | .242 | 40.4 | 42.7 |
| M7078/3-18-4 | ↓ | .007 | .266 | 51.4 | 54.3 |
| M7078/3-18-5 | ↓ | Braid | .289 | 61.9 | 64.5 |
| M7078/3-18-6 | ↓ | ↓ | .317 | 73.0 | 76.0 |
| M7078/3-18-7 | ↓ | ↓ | .317 | 81.8 | 85.2 |
| M7078/3-16-1 | 36 | .005 | .142 | 18.6 | 20.0 |
| M7078/3-16-2 | ↓ | .007 | .248 | 34.9 | 37.0 |
| M7078/3-16-3 | ↓ | .007 | .264 | 48.3 | 51.0 |
| M7078/3-16-4 | ↓ | Braid | .287 | 61.0 | 63.5 |
| M7078/3-16-5 | ↓ | ↓ | .317 | 74.4 | 77.5 |
| M7078/3-16-6 | ↓ | ↓ | .347 | 87.8 | 91.5 |
| M7078/3-16-7 | ↓ | ↓ | .347 | 98.8 | 102.9 |



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TABLE I (Continued)

| Cable part no. | Gage of shield strands (AWG) | Thickness of extruded jacket 1/ (in.)(min) | Major diameter of shielded jacketed cable (in.)(max) | Weight of shielded jacketed cable (lb/1000 ft) | |
|---|---|---|---|---|---|
| | | | | (nom.) 2/ | (max) |
| M7078/3-14-1 | 36 | .005 | .162 | 25.4 | 27.1 |
| M7078/3-14-2 | ↓ | .007 | .288 | 49.0 | 51.6 |
| M7078/3-14-3 | ↓ | Braid | .304 | 67.8 | 70.4 |
| M7078/3-14-4 | ↓ | ↓ | .336 | 86.9 | 90.1 |
| M7078/3-14-5 | ↓ | ↓ | .371 | 106.6 | 111.0 |
| M7078/3-14-6 | ↓ | ↓ | .407 | 126.4 | 131.7 |
| M7078/3-14-7 | ↓ | ↓ | .407 | 143.1 | 149.1 |
| M7078/3-12-1 | 36 | .006 | .184 | 34.8 | 37.3 |
| M7078/3-12-2 | ↓ | Braid | .325 | 67.2 | 70.2 |
| M7078/3-12-3 | ↓ | ↓ | .347 | 94.9 | 98.9 |
| M7078/3-12-4 | ↓ | ↓ | .384 | 124.7 | 129.2 |
| M7078/3-12-5 | ↓ | ↓ | .425 | 151.7 | 158.0 |
| M7078/3-12-6 | ↓ | ↓ | .467 | 180.6 | 188.1 |
| M7078/3-12-7 | ↓ | ↓ | .467 | 205.6 | 214.2 |
| M7078/3-10-1 | 36 | .006 | .206 | 48.7 | 51.3 |
| M7078/3-10-2 | ↓ | Braid | .369 | 93.8 | 97.7 |
| M7078/3-10-3 | ↓ | ↓ | .394 | 134.4 | 140.0 |
| M7078/3-10-4 | ↓ | ↓ | .437 | 174.3 | 181.6 |
| M7078/3-10-5 | ↓ | ↓ | .484 | 214.5 | 223.4 |
| M7078/3-10-6 | ↓ | ↓ | .533 | 256.2 | 266.9 |
| M7078/3-10-7 | ↓ | ↓ | .533 | 293.1 | 305.3 |

1/ When thickness of extruded jacket is specified in the above table, the cable shall be made with an extruded jacket. When braid is indicated, the cable shall be made with an underlying tape and a braid jacket.

2/ Nominal values for weight of shielded jacketed cable are given for information only. Nominal values are not requirements.

SUPERSESSION DATA: This specification sheet includes the requirements for the "A" dash number cables of MS25313. The individual cables of the specification sheet replace and supersede the "A" dash number cables of corresponding basic wire size and quantity of conductors in MS25313. Examples: Part number M7078/3-22-1 cable of this specification sheet replaces and supersedes dash number -A221 cable of MS25313; part number M7078/3-20-7 cable of the specification sheet replaces and supersedes dash number -A207 cable of MS25313.



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